

1 Status information on the order

		
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This document describes the current status information and actual data pertaining to an order. Basically, this data results from the operation-related data collection in BDE (Shop Floor Data Collection). The document [Data structure of order](#) describes basic operation data.

The below-mentioned data is available in certain applications. The categories used there might deviate from the categories mentioned below. It is not a fault if a piece of information is missing in a report/evaluation.

Status

Order status (LED and text)

The order status represents the status of a whole production order. Possible statuses are:

-  prepared: the order has not yet been started.
-  started: the order has already been started.
-  finished: the order has been finished.

Status since

Time since the order status exists.

Progress

Number of OPs

Total of operations for which you can enter and post data, i.e. operations with an assigned processing code that is set to the "recordable" option.

Finished OPs

Total of the finished, recordable operations, i.e. operations for which you can record and post data and that are assigned the status "finished" or "archived".

Remaining run time

Total of remaining run times (RRT formula 1) of all recordable operations pertaining to the order that have not yet been finished. Remaining run times that return negative results due to the formula calculation are processed as "0".

Remaining run time 2

Total of remaining run times (RRT formula 2) of all recordable operations pertaining to the order that have not yet been finished. Remaining run times that return negative results due to the formula calculation are processed as "0".

Current operation

The operation that is logged in with the highest operation number within the order network of the current order.

Current OP sequence

Reserved.

Current OP number

Reserved.

Current OP split

Reserved.

Current OP name

Reserved.

Current target quantity of OP

Reserved.

Current OP unit

Reserved.

Current yield of OP

Reserved.

Quantities

Target quantity

Base quantity from the order header. This quantity and all quantities that follow are entered in base quantity unit!

Target scrap

Base quantity from order header

Unit

Base quantity unit from the order header

Yield

Recorded yield in base quantity unit of the last operation that can be recorded/posted for the order.

It is not the "current operation", as this quantity is currently not significant since the order itself has not yet delivered a quantity when production is still running and only half of the OPs have been processed at that point in time.

A program integrated in the HYDRA Scheduler identifies the last recordable operation of the order at regular intervals (by default: once a day). You can find further information on how to activate this program [here](#).



To do so, you have to store the conversion factors from primary quantity unit to base quantity unit in the operation. The same also applies to scrap, rework and open quantity.

Scrap

Sum of the scrap quantities entered for all operations posted in base quantity unit.

Irrespective of whether data can be entered/posted or if the OP is *active/inactive*.

Rework

Sum of the rework quantities entered for all operations posted in base quantity unit.

Open quantity

Sum of the open quantities entered for all operations posted in base quantity unit.

Target times

Planned lead time

HYDRA [Scheduling](#), if activated, calculates the planned lead time. You can calculate the planned lead time from the difference between the scheduled end and the scheduled beginning of the order, compared to the Gregorian calendar.

Minimum lead time

HYDRA [Scheduling](#), if activated, calculates the minimum lead time of the order. The minimum lead time results from the difference between the planned lead time and the time that can be reduced (see below).

Order buffer

The scheduling each results in the buffer of the order relating to the required date. The basic end date stands for the required date. You can determine the order buffer from the difference between the basic end date and the scheduled end date. Once this buffer gets negative, the order is delayed.

OP buffer

HYDRA [Scheduling](#), if activated, calculates the OP buffer. This buffer results from the total operation buffers of the individual operations pertaining to the order. You can calculate the operation buffer from the difference between the latest start date (LSD) and the earliest start date (ESD) of the operation.

Reducible time

HYDRA [Scheduling](#), if activated, calculates the reducible time. This time represents the sum total of reducible times affecting all operations pertaining to the order. This value indicates how many (more) hours can be reduced from the lead time of an order.

Target setup time

Total of setup time, additional setup time and retooling time of all active operations.

Target processing time

Total of the processing time of all active operations.

Target execution time

Total of planned setup time and planned processing time.

Personnel deployment/labor utilization

Total target labor utilization of all recordable, active operations.

Delivery time

Total delivery time of all recordable, active OPs assigned to the "external processing" option.

Waiting time

Total waiting time of all recordable, active OPs.

Wait time

Total wait time/idle time of all recordable, active OPs.

Transport time

Total transport time of all recordable, active OPs.

Actual times**Retention period of order**

The retention period of the order results from the period of time between:

- when the order was first transferred from the PPS system ("order release" = creation date of the order header in HYDRA) and
- when the last active operation of the order is actually logged off (chronological order). The retention period of the order is 0 as long as the order has not yet been finished.

Please note:

We cannot identify or take into account if

- the PPS system transfers the order for the first time or
- the order has meanwhile been deleted and resent due to technical problems.

In case an order is transferred several times and the previous order is deleted with each transfer, the creation date of the order header coincides with the last data transfer (chronological order).

Lead time

The order duration results from the period of time between:

- the first logon of an operation of the order and
- the logoff of the last active operation (chronological order). The lead time is 0 as long as the order has not yet been finished.

Setup time

The setup time of the order is based on the total setup times (RPA 7) of all active operations that can be posted.

Processing time

The processing time of the order is based on the total main production times (RPA 11) of all active operations that can be posted.

Downtime

The downtime of the order is based on the total downtimes (RPA 1 to 6, RPA 8 to 10) of all active operations that can be posted.

Occupancy time

The occupancy/assignment time is based on the total setup times (RPA 7), processing times (RPA 11) and downtimes (RPA 1...6, RPA 8...10) of all active operations that can be posted.

Personnel deployment/labor utilization

Total labor utilization of all active operations of the order that can be posted.

**Transition time **

The transition time refers to the total transition times of all operations.

The transition time of an operation refers to the time period between

- the (last) actual logoff of the current ("preceding") operation and
- the (first) actual logon of the next ("subsequent") operation. The transition time is 0 if two OPs coincide.

Times are synchronized based on the Gregorian calendar (the shift model is not taken into account).

Please note: According to technical literature, transition times often represent the largest share of the lead time. In contrast to this, the occupancy/assignment time often only amounts to approximately 10-20%.

**Wait time/idle time **

The order-related wait time results from times during which no operation of the order was logged on. Consequently, this time includes the operation-related wait times. You can identify the "gaps" using the BDE log records with record type U/E.

Times are synchronized based on the Gregorian calendar (the shift model is not taken into account).



The total wait time of all operations does not necessarily have to correspond to the order's wait time. This is particularly true if you process operations simultaneously.

Personnel deployment/labor utilization

The labor utilization of the order is the total of personnel deployment times of all active operations that can be posted.

RPA**SUT...BKS**

Total of the times posted to the relevant resource performance account of all active operations.

Actual dates**Order start date/time**

Results from the start of the earliest operation.

Order end date/time

Results from the end of the last operation logged off (chronological order). Empty, as long as the order has not been finished/archived.

Deviations**Yield/target quantity [%]**

$100 / \text{Target quantity} * \text{yield}$

Setup [%]

Total $(100 / \text{target setup} * \text{RPA7})$ of all finished operations that can be posted.

Processing [%]

Total $(100 / \text{target processing} * \text{RPA11})$ of all finished operations that can be posted.

Labor utilization [%]

Total $(100 / \text{target labor utilization} * \text{labor utilization})$ of all finished operations that can be posted.

KPIs

Displayed in % including 2 decimal places

Utilization efficiency (rate of capacity utilization)

Relation of the processing time (RPA 11) compared to the assignment/occupancy time (RPA 1...11) in percent.

Setup ratio

Relation of the setup time (RPA 7) compared to the assignment/occupancy time (RPA 1..11) in percent.

Throughput rate

Produced quantity/lead time (DLZ)

Formula: yield.primary / DLZAU

DLZ= (order end (logoff of last OP) – order start (first logon of first OP of order))

Use the formula **trp_au** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Production process ratio

Main production time/lead time (DLZ)

Formula: rpa11/ DLZAU * 100

DLZAU= (order end (logoff of last OP) – order start (first logon of first OP of order))

Use the formula **ppr_au** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Actual to planned scrap ratio

Scrap quantity/planned scrap quantity

Formula: scrap.primary / order.plan.scrap.base *100

Use the formula **psc_au** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Scrap ratio

Scrap quantity/ produced quantity

Formula: scrap.primary / (yield.primary + scrap.primary + rework.primary + problem.primary) * 100

Use the formula **scr_au** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Rework ratio

Rework quantity / produced quantity

Formula: rework.primary / (yield.primary + scrap.primary + rework.primary + problem.primary) * 100

Use the formula **rew_au** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Allocation ratio

Occupancy time / lead time

Formula: BLZ / DLZAU * 100

BLZ = total occupancy/assignment time = RPA1-11

DLZAU= (order end (logoff of last OP) – order start (first logon of first OP of order))

Use the formula **all_au** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.



You cannot change the formula elements DLZAU and/or DLZAG.



The KPIs *throughput rate*, *production process ratio*, *actual to planned scrap ratio*, *scrap rate*, *rework ratio* and *allocation ratio* are only available if you enable the upgrade [orst82](#).

Information on order-related KPIs

When calculating order-related KPIs (in this document identified with the icon ), the system integrates all operations of an order matching the below-mentioned criteria:

- You can enter/post data for the operation ("recordable" option at the operation)
- It is a "normal" operation or a split master (individual splits are not taken into account).
- The operation pertains to an active sequence.
- The operation is neither inactive nor deleted (control indicators <> "Y" and <> "D").

Merged operations are not taken into account.



A program integrated in the HYDRA Scheduler calculates order-related KPIs (identified by ) at regular intervals (by default: once a day). You can find further information on how to activate this program [here](#).